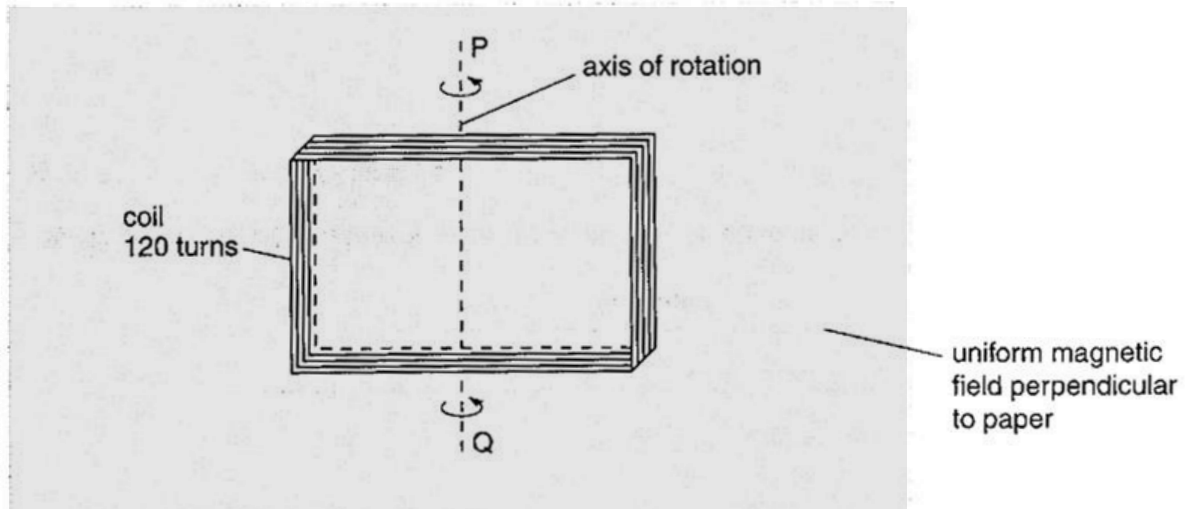


**26**

The coil in a generator is situated in a uniform magnetic field, as shown.

**[Turn over**



The coil of area  $A$  is rotated about the axis PQ with an angular speed  $\omega$ .

Which of the following changes to  $A$  and  $\omega$  increases the electromotive force (e.m.f.) in the coil?

|  |          | $A$      | $\omega$ |  |
|--|----------|----------|----------|--|
|  | <b>A</b> | increase | increase |  |
|  | <b>B</b> | increase | decrease |  |
|  | <b>C</b> | decrease | increase |  |
|  | <b>D</b> | decrease | decrease |  |

**Ans: A**

increasing  $A$  increases maximum magnitude flux linkage

which increases rate of change of magnetic flux linkage

increasing  $\omega$  increases rate of change of magnitude flux linkage